Smart Crop Recommendation

Farm Analysis & Decision Support System

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System Framework Pipeline

Collection

Cleaning

Feature Eng.

ML Training

Evaluation

Deployment

A robust 7-stage architecture designed for high-accuracy agricultural intelligence.

Data Collection & Ingestion

Unified Data Sources

We leverage a multi-modal approach to capture a holistic view of the farm environment.

- **Kaggle Dataset:** Sourced via CSV for historical baseline records.
- **IoT Sensors:** Real-time soil telemetry.

Dataset Feature Specifications

Primary features extracted from our **Kaggle CSV Repository** for model training:

Feature	Metric Type	Role in Prediction
N-P-K Ratios	Soil Macronutrients	Determines nutritional viability.
pH Level	Soil Acidity	Affects mineral absorption rates.
Humidity & Temp	Climate Data	Predicts transpiration and growth cycles.
Rainfall	Weather API	Calculates hydration requirements.

Cleaning & Feature Engineering

Data Cleaning

Removing noise and handling missing values from the Kaggle CSV to ensure high data integrity.

Standardization

Normalizing environmental variables like Temperature and Rainfall to a uniform scale for the models.

Feature Selection

Identifying the most influential factors (NPK, pH) that drive crop suitability and yield optimization.

Model Training & Logic

Random Forest

Ensemble Power: Combines multiple decision trees to handle non-linear patterns with 99%+ accuracy.

KNN

Proximity Logic: Predicts crops based on the similarity of soil features to existing data clusters.

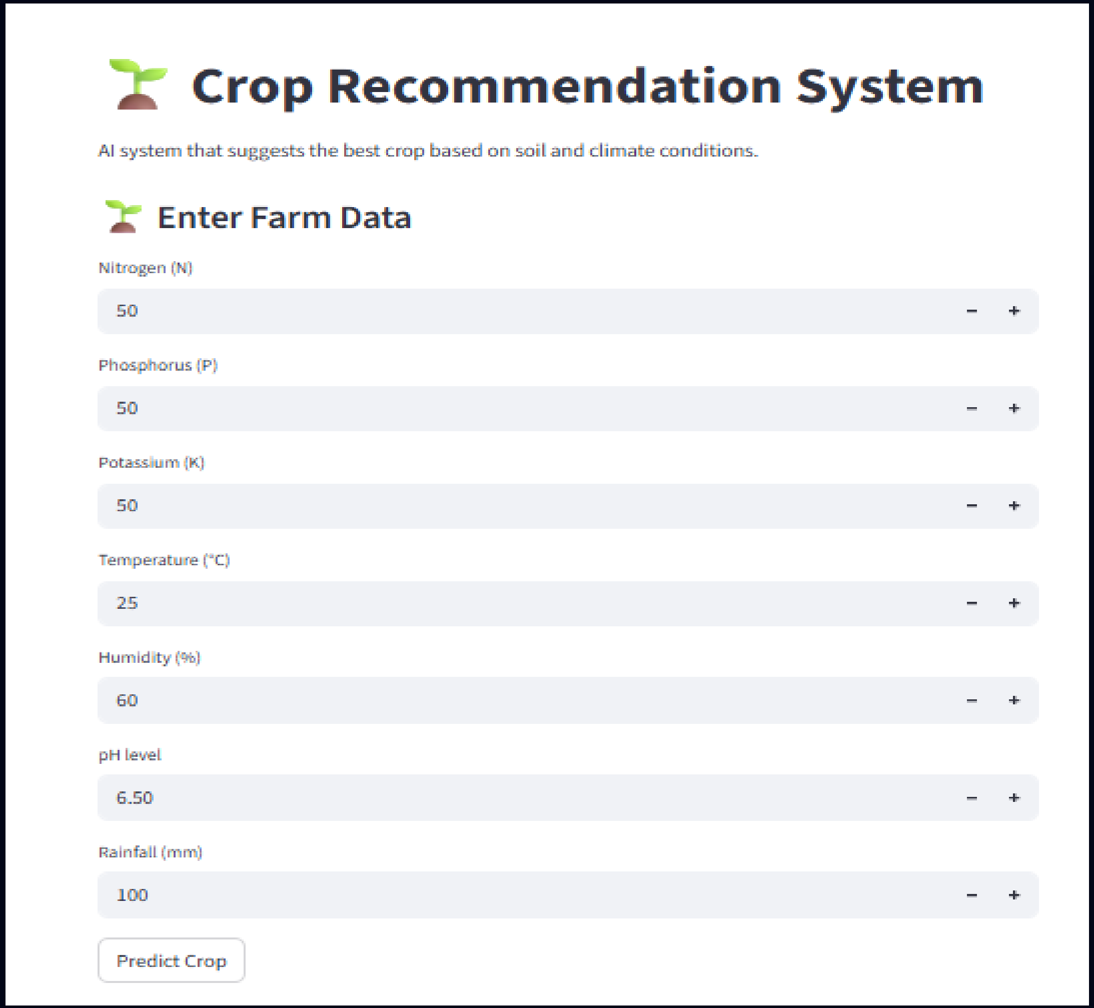
Linear Regression

Trend Mapping: Establishes direct relationships between soil nutrients and growth expectations.

Model Accuracy Benchmarks

Random Forest provides the most reliable predictions for diverse regional soil conditions.

System Deployment Architecture



Frontend: React

Dynamic and responsive dashboard for farmers.

Backend: Flask

Robust REST API connecting models to the interface.

Insight: Streamlit

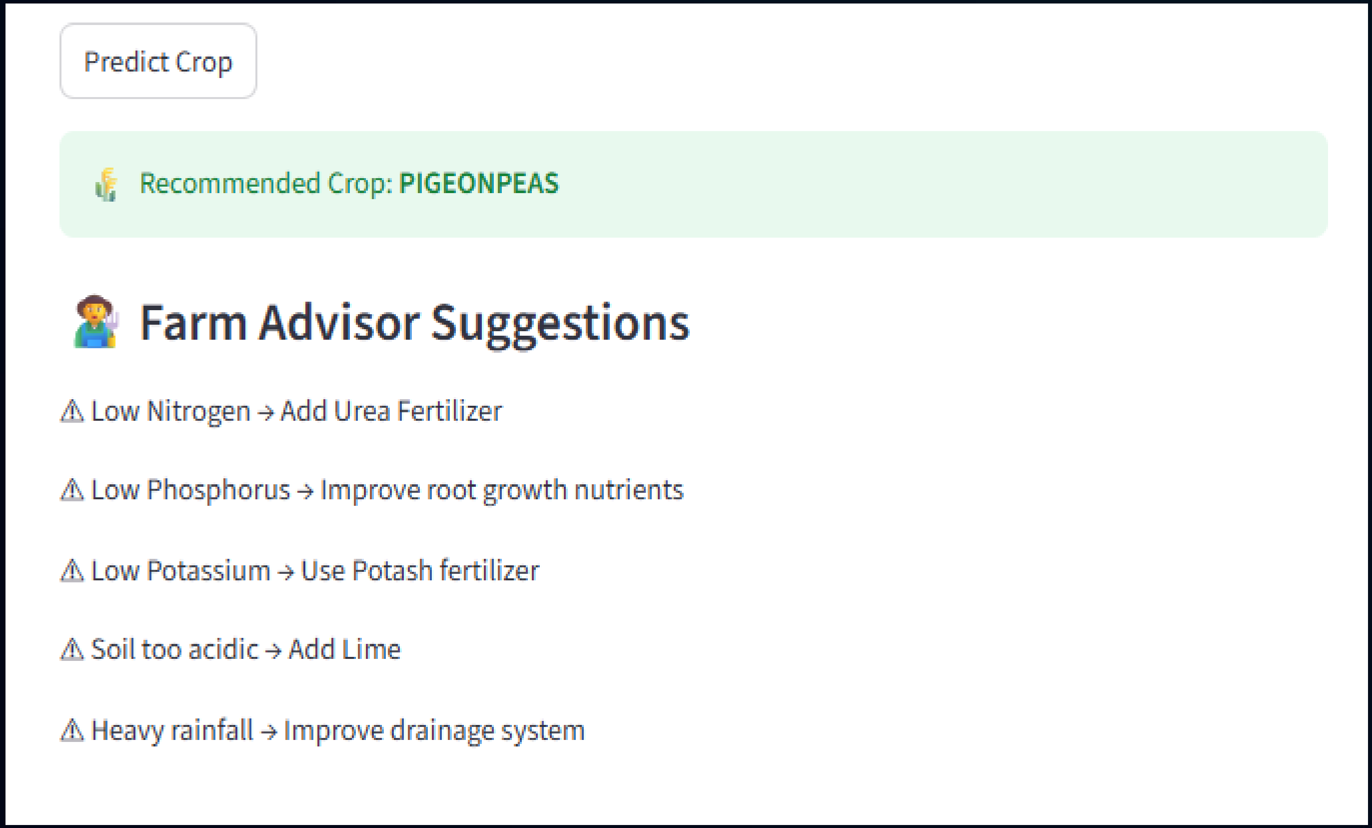
Real-time analytical visualization and mobile deployment.

LIVE UI SHOWCASE

Streamlit Deployment

A mobile-responsive dashboard designed for immediate farm-side deployment and real-time crop analysis.

https://gvgbhwnwym6hpnwk9rptka.streamlit.app/



Questions?

Thank you for your time and attention.